

pkt
LOG

Syslog Ingest & Management Platform

Platform Overview · v1.0

What is pktLog?

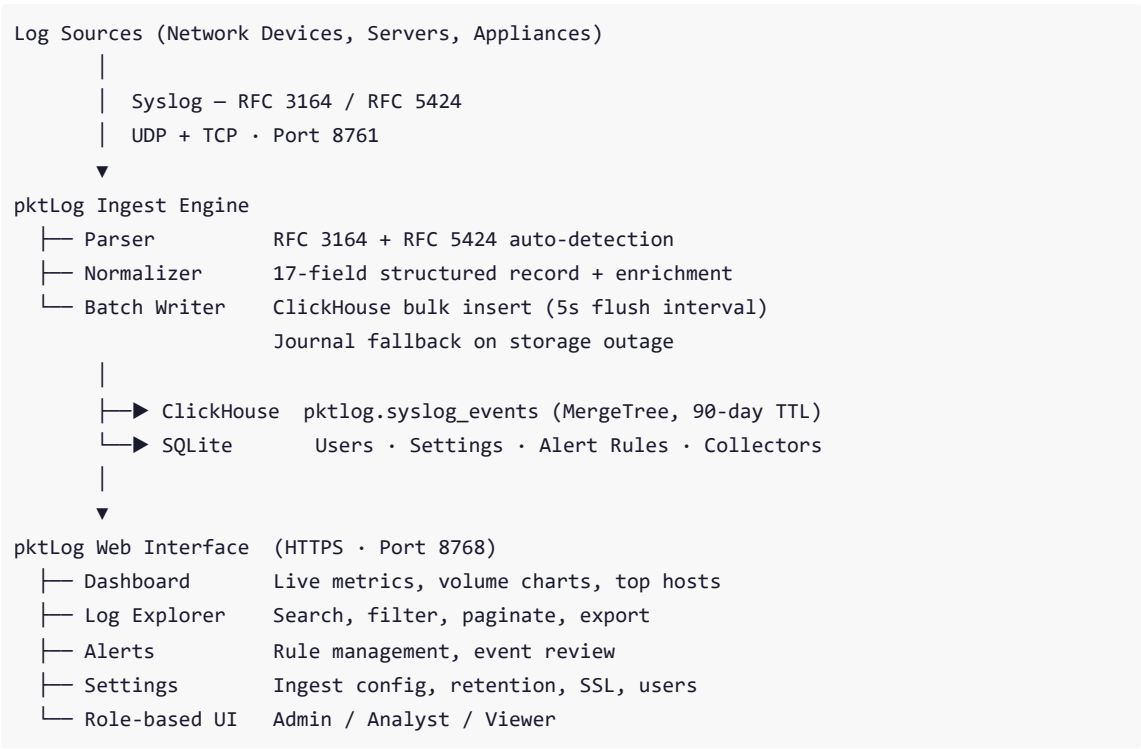
pktLog is a centralized syslog collection, storage, and analysis platform. It ingests syslog data from any RFC 3164 or RFC 5424 compliant source — network devices, servers, applications, and security appliances — normalizes every message into structured fields, and stores them in a high-performance columnar database for real-time search and analytics.

The platform is accessed through a secure web interface that provides role-based dashboards, a full-featured log explorer, alert management, and configuration tools — all in a single integrated application.

Key Capabilities

Capability	Details
Ingest Protocols	UDP and TCP on a single configurable port
Syslog Standards	RFC 3164 (BSD) and RFC 5424 (structured)
Normalized Fields	17 fields per event including severity, facility, host, program, pid, timestamps
Storage Engine	ClickHouse — columnar, high-volume, configurable TTL
Web Interface	React + TypeScript dashboard served over HTTPS
Authentication	JWT with configurable lifetime, bcrypt password hashing
Access Control	Three roles: Admin, Analyst, Viewer
Alert Engine	Rule-based alerting on severity, silence, rate spikes, patterns
Export	CSV and JSON export from the Log Explorer (Analyst and Admin)
Multi-Site	syslog-ng collectors with disk-buffer at remote sites

Data Flow Summary



User Roles

Admin

Full platform control. Manages users, configuration, ingest settings, alert rules, SSL certificates, and system maintenance. Should be limited to platform operators.

Analyst

Read and write access for log investigation work. Can search all events, export results, acknowledge and resolve alerts. Cannot modify platform configuration.

Viewer

Read-only access to the dashboard and log browser. Suitable for operations staff who need visibility without investigation or configuration rights.

Technology Stack

Component	Technology
Backend	FastAPI (Python) — async, single systemd service
Log Storage	ClickHouse — columnar OLAP, sub-second aggregations
App Database	SQLite — users, settings, alerts, collector registry

Frontend	React 18 + TypeScript + Vite + Tailwind CSS
Auth	JWT (python-jose) + bcrypt (passlib)
HTTPS	Auto-detected TLS — upload cert via Settings UI
Collectors	syslog-ng + disk-buffer at remote sites

For detailed architecture diagrams, API reference, and deployment guidance, refer to the [Architecture document](#). For role-specific usage instructions, refer to the [Admin](#), [Analyst](#), or [Viewer guides](#).